

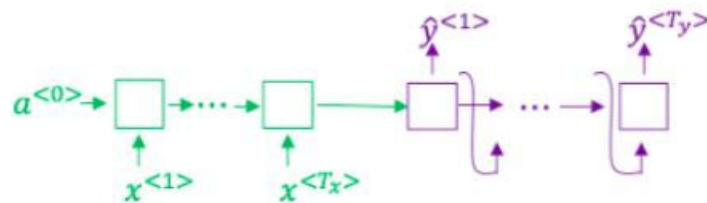
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1. Consider using this encoder-decoder model for machine translation.

1 / 1 point



True/False: This model is a “conditional language model” in the sense that the encoder portion (shown in green) is modeling the probability of the input sentence x .

- ☐ True
- ☒ False

✓ **Correct**

The encoder-decoder model for machine translation models the probability of the output sentence y conditioned on the input sentence x . The encoder portion is shown in green, while the decoder portion is shown in purple.

2. In beam search, if you increase the beam width B , which of the following would you expect to be true?

- ☐ Beam search will use up less memory.
- ☒ Beam search will generally find better solutions (i.e. do a better job maximizing $P(y|x)$).
- ☐ Beam search will converge after fewer steps.
- ☐ Beam search will run more quickly.

✓ **Correct**

As the beam width increases, beam search runs more slowly, uses up more memory, and converges after more steps, but generally finds better solutions.

3. True/False: In machine translation, if we carry out beam search using sentence normalization, the algorithm will tend to output overly short translations.

1 / 1 point

- ☐ True
- ☒ False

✓ **Correct**

In machine translation, if we carry out beam search without using sentence normalization, the algorithm will tend to output overly short translations.

4. Suppose you are building a speech recognition system, which uses an RNN model to map from audio clip x to a text transcript y . Your algorithm uses beam search to try to find the value of y that maximizes $P(y \mid x)$.

On a dev set example, given an input audio clip, your algorithm outputs the transcript $\hat{y} = \text{"I'm building an A Eye system in Silly con Valley."}$, whereas a human gives a much superior transcript $y^* = \text{"I'm building an AI system in Silicon Valley."}$

According to your model,

$$P(\hat{y} \mid x) = 7.21 \times 10^{-8}$$

$$P(y^* \mid x) = 1.09 \times 10^{-7}$$

Would you expect increasing the beam width B to help correct this example?

- ☒ Yes, because $P(y^* \mid x) > P(\hat{y} \mid x)$ indicates the error should be attributed to the search algorithm rather than to the RNN.
- ☐ No, because $P(y^* \mid x) > P(\hat{y} \mid x)$ indicates the error should be attributed to the RNN rather than to the search algorithm.
- ☐ Yes, because $P(y^* \mid x) > P(\hat{y} \mid x)$ indicates the error should be attributed to the RNN rather than to the search algorithm.
- ☐ No, because $P(y^* \mid x) > P(\hat{y} \mid x)$ indicates the error should be attributed to the search algorithm rather than the RNN.

✔ **Correct**

$P(y^* \mid x) > P(\hat{y} \mid x)$ indicates the error should be attributed to the search algorithm rather than to the RNN. Increasing the beam width will generally allow beam search to find better solutions.

1 / 1 point

5. Continuing the example from Q4, suppose you work on your algorithm for a few more weeks, and now find that for the vast majority of examples on which your algorithm makes a mistake, $P(y^* | x) > P(\hat{y} | x)$. This suggests you should focus your attention on improving the RNN.

☒ False

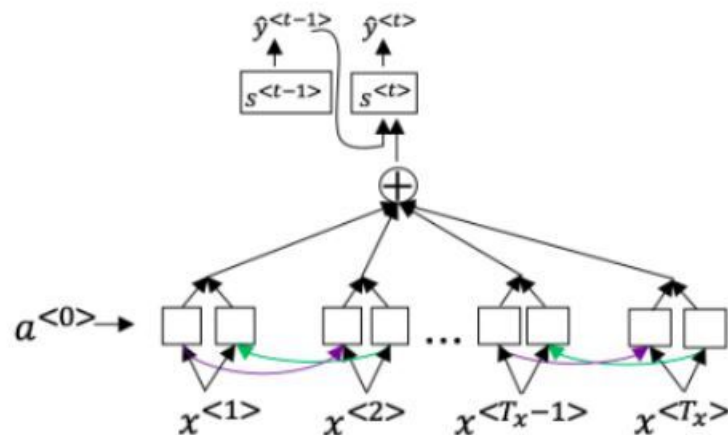
☐ True

✓ **Correct**

$P(y^* | x) > P(\hat{y} | x)$ indicates the error should be attributed to the search algorithm rather than to the RNN.

6. Consider the attention model for machine translation.

0.5 / 1 point



Further, here is the formula for $\alpha^{<t,t'>}$.

$$\alpha^{<t,t'>} = \frac{\exp(e^{<t,t'>})}{\sum_{t'=1}^{T_x} \exp(e^{<t,t'>})}$$

Which of the following statements about $\alpha^{<t,t'>}$ are true? Check all that apply.

☐ $\sum_t \alpha^{<t,t'>} = 1$ (Note the summation is over t .)

☒ $\sum_{t'} \alpha^{<t,t'>} = 1$ (Note the summation is over t' .)

✔ Correct

Correct! If we sum over $\alpha^{<t,t'>}$ for all t' (the formulation can be seen in the image), the numerator will be equal to the denominator, therefore, $\sum_{t'} \alpha^{<t,t'>} = 1$.

☐ We expect $\alpha^{<t,t'>}$ to be generally larger for values of $a^{<t'>}$ that are highly relevant to the value the network should output for $y^{<t>}$. (Note the indices in the superscripts.)

☒ We expect $\alpha^{<t,t'>}$ to be generally larger for values of $a^{<t>}$ that are highly relevant to the value the network should output for $y^{<t'>}$. (Note the indices in the superscripts.)

✘ This should not be selected

Incorrect! We expect $\alpha^{<t,t'>}$ to be generally larger for values of $a^{<t'>}$ that are highly relevant to the value the network should output for $y^{<t>}$. $\alpha^{<t,t'>}$ is equal to the amount of attention $y^{<t>}$ should pay to $a^{<t'>}$. So, if a value of $a^{<t'>}$ is highly relevant to $y^{<t>}$, then the attention coefficient

7. The network learns where to “pay attention” by learning the values $e^{\langle t, t' \rangle}$, which are computed using a small neural network:

1 / 1 point

We can't replace $s^{\langle t-1 \rangle}$ with $s^{\langle t \rangle}$ as an input to this neural network. This is because $s^{\langle t \rangle}$ depends on $\alpha^{\langle t, t' \rangle}$ which in turn depends on $e^{\langle t, t' \rangle}$; so at the time we need to evaluate this network, we haven't computed $s^{\langle t \rangle}$ yet.

☒ True

☐ False

✓ Correct

8. Compared to the encoder-decoder model shown in Question 1 of this quiz (which does not use an attention mechanism), we expect the attention model to have the greatest advantage when:

1 / 1 point

☒ The input sequence length T_x is large.

☐ The input sequence length T_x is small.

✓ Correct

9. Under the CTC model, identical repeated characters not separated by the “blank” character (`_`) are collapsed. Under the CTC model, what does the following string collapse to?

aaa_aaaaaa_____rr_dddddddddd_____v_aaaaaa_rrrr_____kk

- ☐ aaaaaaaaaarrdddddddddvaaaaaarrrkk
- ☐ ardvark
- ☐ aa rd var k
- ☒ aardvark

✔ **Correct**

The basic rule for the CTC cost function is to collapse repeated characters not separated by "blank". If a character is repeated, but separated by a “blank”, it is included in the string.

10. In trigger word detection, if the target label for $x^{<t>}$ is 1:

1 / 1 point

- ☒ Someone has just finished saying the trigger word at time t .
- ☐ There is exactly one trigger word.
- ☐ Only one word has been stated.
- ☐ The total time that the trigger word detection algorithm has been running is 1.

✔ **Correct**

Target labels indicate whether or not a trigger word has been said.